Lecture Notes 3

Econometrics

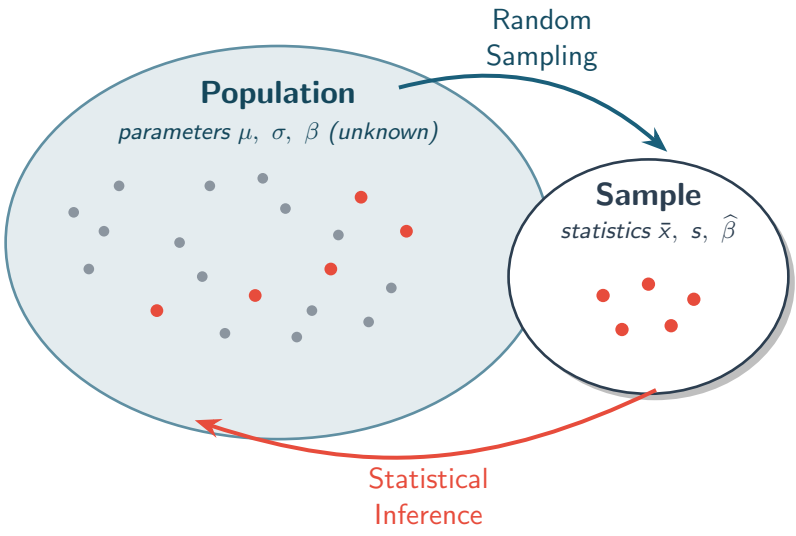
Inference

Markeljan Blacëri

Department of Economics and Management
University of Padova
markeljan.blaceri@phd.unipd.it

A.Y. 2026-2027

What is Inference?



Sampling Distributions of the OLS Estimators

- Knowing the expected value and variance of the OLS estimators is useful for describing the precision of the OLS estimators.
- However, in order to perform statistical inference, we need to know the full sampling distribution of the $\widehat{\beta}_j$.
- We assume that the unobserved error is **normally distributed** in the population.

Normality Assumption

The population error ε is independent of the explanatory variables $x_1, x_2, \ldots, x_k$ and is normally distributed with zero mean and variance σ^2.

$$\varepsilon \sim \mathcal{N}(0, \sigma^2)$$

Sampling Distributions of the OLS Estimators

Normality of the error term translates into normal sampling distributions of the OLS estimators.

Normal Sampling Distributions

Conditional on the sample values of the independent variables

$$\widehat{\beta}_j \sim \mathcal{N}\left(\beta_j, \text{Var}(\widehat{\beta}_j)\right)$$

Therefore

$$\frac{\widehat{\beta}_j - \beta_j}{sd(\widehat{\beta}_j)} \sim \mathcal{N}(0,1)$$

What If the Errors Are Not Normal?

- The normality assumption $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ is **strong** — and often unrealistic (e.g. wages, which are skewed and non-negative).
- The good news: we do not really need it. Even with non-normal errors, OLS remains well-behaved in **large samples**.
- **Consistency.** Under the zero conditional mean assumption, OLS converges in probability to the truth:

$$\widehat{\beta}_j \xrightarrow{p} \beta_j \qquad \text{as } n \to \infty.$$

  As the sample grows, sampling error vanishes and $\widehat{\beta}_j$ gets arbitrarily close to β_j.
- **Asymptotic normality.** By the Central Limit Theorem, the sampling distribution of $\widehat{\beta}_j$ is **approximately normal** in large samples — *regardless* of the distribution of ε:

$$\frac{\widehat{\beta}_j - \beta_j}{se(\widehat{\beta}_j)} \xrightarrow{d} \mathcal{N}(0, 1).$$

Testing Hypotheses about a Single Population Parameter

- The population model can be written as

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + \varepsilon$$

- The β_j are unknown features of the population.
- We can hypothesize about the value of β_j and then use statistical inference to test our hypothesis.
- To construct hypotheses tests, we need the following result:

t Distribution for the Standardized Estimators

$$\frac{\widehat{\beta}_j - \beta_j}{se(\widehat{\beta}_j)} \sim t_{n-k-1}$$

- The t distribution comes from the fact that the constant σ in $sd(\widehat{\beta}_j)$ has been replaced with the random variable $\widehat{\sigma}$.

Testing Hypotheses about a Single Population Parameter

- Our primary interest lies in testing the null hypothesis

$$H_0 : \beta_j = 0$$

- Because β_j measures the partial effect of x_j on y, after controlling for all other independent variables, means that, once $x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_k$ have been accounted for, x_j has no effect on the expected value of y.
- The statistic we use to test the null hypothesis is called the t statistic or the t ratio of $\widehat{\beta}_j$.

$$t_{\widehat{\beta}_j} \equiv \frac{\widehat{\beta}_j}{se(\widehat{\beta}_j)}$$

Example: Wage Equation

Consider the wage equation

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + \varepsilon$$

The null hypothesis $H_0 : \beta_2 = 0$ means that, once education and tenure have been accounted for, the number of years in the workforce (*exper*) has no effect on hourly wage.

The Significance Level and Two Types of Error

Because we decide using a *sample*, our conclusion about H_0 can be wrong in two ways:

	H_0 **true**	H_0 **false**
Fail to reject H_0	Correct	Type II error
Reject H_0	Type I error	Correct (power)

- A **Type I error** is rejecting H_0 when it is in fact true. We control its probability by fixing the **significance level**:

$$\alpha = P\big(\text{reject } H_0 \mid H_0 \text{ true}\big).$$

- Common choices are $\alpha = 0.01,\ 0.05,\ 0.10$. A 5% level means we accept wrongly rejecting a true H_0 in 5% of samples.
- A **Type II error** is failing to reject H_0 when it is false; the **power** of the test is the probability of correctly rejecting a false H_0.
- Lowering α guards against Type I errors but raises the chance of a Type II error — there is a **trade-off**.

Testing against One-Sided Alternatives

- To determine a rule for rejecting H_0, we need to decide on the relevant alternative hypothesis.
- Consider a one-sided alternative of the form:

$$H_1 : \beta_j > 0$$

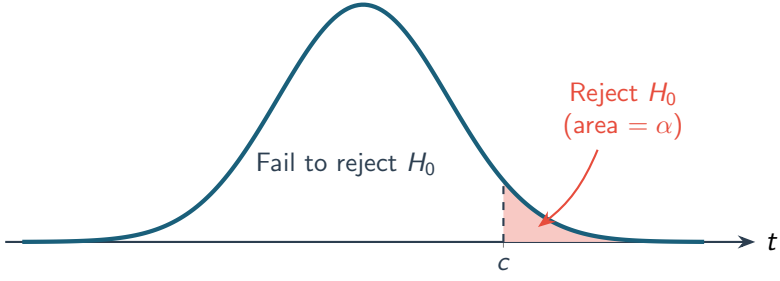
- When we state the alternative as this, we are really saying that the null hypothesis is $H_0 : \beta_j \leq 0$.
- Decide on a significance level or the probability of rejecting H_0 when it is in fact true.
  - Suppose we have decided on a 5% significance level.
  - This means we are willing to mistakenly reject H_0 when it is true 5% of the time.

The Rejection Rule when $H_1 : \beta_j > 0$

The rejection rule is that H_0 is rejected in favor of H_1 at the α% significance level if

$$t_{\widehat{\beta}_j} > c$$

where c is the critical value, chosen so that $P(t_{\widehat{\beta}_j} > c) = \alpha$ under H_0.



Example: Wage Equation

The estimated hourly wage equation is

$$\widehat{\log(wage)} = \underset{(0.104)}{0.284} + \underset{(0.007)}{0.092}\,educ + \underset{(0.0017)}{0.0041}\,exper + \underset{(0.003)}{0.022}\,tenure$$

Test whether the return to *exper*, controlling for *educ* and *tenure*, is zero in the population, against the alternative that it is positive. The sample size is $n = 526$, thus use the standard normal critical values. The 5% critical value is 1.645.

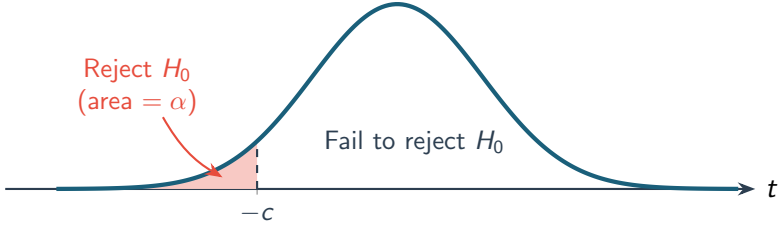
Testing against One-Sided Alternatives

- Another one-sided alternative hypothesis that arises in applications is that the parameter is less than zero

$$H_1 : \beta_j < 0$$

- The rejection rule is

$$t_{\hat{\beta}_j} < -c$$



Example: Wage Equation with Gender Dummy

The estimated hourly wage equation is

$$\widehat{\log(wage)} = \underset{(0.102)}{0.501} + \underset{(0.007)}{0.087}\, educ + \underset{(0.002)}{0.005}\, exper + \underset{(0.003)}{0.017}\, tenure - \underset{(0.037)}{0.301}\, female$$

where $female = 1$ for women and 0 for men. Test whether, controlling for *educ*, *exper*, and *tenure*, there is a wage penalty for women:

$$H_0 : \beta_{female} = 0 \quad \text{against} \quad H_1 : \beta_{female} < 0.$$

The sample size is $n = 526$, so use the standard normal critical values. The 5% critical value is 1.645.

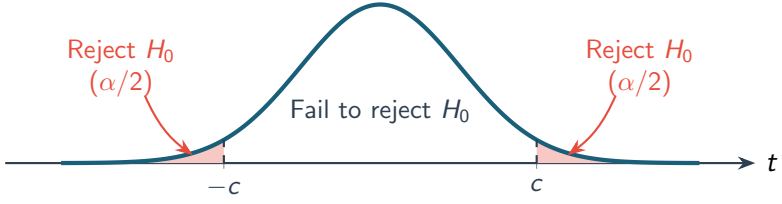
Two-Sided Alternatives

- It is common to test the null hypothesis $H_0 : \beta_j = 0$ against a two-sided alternative:

$$H_1 : \beta_j \neq 0$$

- Under this alternative, x_j has a ceteris paribus effect on y without specifying whether the effect is positive or negative.
- The rejection rule for $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$ is

$$|t_{\hat{\beta}_j}| > c$$



Testing Other Hypotheses about β_j

- We sometimes want to test whether β_j is equal to some other given constant.
- The null is stated as

$$H_0 : \beta_j = a_j$$

  where a_j is our hypothesized value of β_j.
- The appropriate t statistic is

$$t = \frac{\widehat{\beta}_j - a_j}{se(\widehat{\beta}_j)} \overset{H_0}{\sim} t_{n-k-1}$$

Example: Housing Prices and Air Pollution

For a sample of 506 communities in the Boston area, we estimate a model relating median housing price (*price*) in the community ti various community characteristics: (*nox*) is the amount of nitrogen oxide in the air, in parts per million; *dist* is a weighted distance of the community from five employment centers, in miles; *rooms* is the average number of rooms in houses in the community; and *stratio* is the average student-teacher ratio of schools in the community.

$$\log(price) = \beta_0 + \beta_1 \log(nox) + \beta_2 \log(dist) + \beta_3 rooms + \beta_4 stratio + u$$

Using data, the estimated model is

$$\log(price) = \underset{(0.32)}{11.08} - \underset{((0.117))}{0.954}\log(nox) - \underset{(0.043)}{0.134}\log(dist) + \underset{(0.019)}{0.255}rooms - \underset{(0.006)}{0.052}stratio$$

1. Are all coefficients statistically significant?
2. Test the null hypothesis $H_0 : \beta_1 = -1$.

The p-Value as a Tail Area

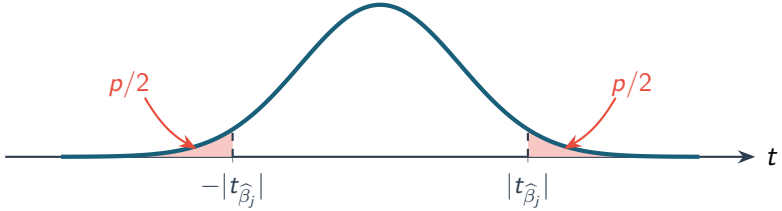
- Rather than testing at different significance levels, it is more informative to answer the following question:

  What is the smallest significance level at which the null hypothesis would be rejected?

- The p-value is the probability, computed *under* H_0, of drawing a t statistic at least as extreme as the one we observed.
- For a two-sided test $H_0 : \beta_j = 0$ vs. $H_1 : \beta_j \neq 0$:

$$p = P\big(|T| > |t_{\widehat{\beta}_j}|\big) = 2\,P\big(T > |t_{\widehat{\beta}_j}|\big),$$

  where $T \sim t_{n-k-1}$ (standard normal in large samples).



Making Decisions with p-Values

- A **small** p-value is evidence *against* H_0; a **large** p-value is weak evidence against it.
- Decision rule: reject H_0 at significance level α if and only if

$$p < \alpha.$$

- This is exactly the critical-value rule in disguise:

$$p < \alpha \iff |t_{\widehat{\beta}_j}| > c.$$

- The p-value is the **observed significance level**: reporting it lets the reader apply *any* α without redoing the test.

Example: Computing a p-Value

Suppose a two-sided test yields $t_{\hat{\beta}_j} = 1.85$ with a large sample, so we use the standard normal.

$$p = 2\,P(Z > 1.85) = 2\,(0.0322) = 0.064.$$

- Since $p = 0.064 > 0.05$, we **fail to reject** H_0 at the 5% level.
- But $p = 0.064 < 0.10$, so we *would* reject at the 10% level.
- Reporting p conveys all of this at once — far more informative than a single reject/fail-to-reject verdict.

Confidence Intervals

- We can easily construct a confidence interval for the population parameter β_j.
- Confidence intervals are also called interval estimates because they provide a range of likely values for the population parameter, and not just a point estimate.
- Using the fact that

$$\frac{\widehat{\beta}_j - \beta_j}{se(\widehat{\beta}_j)} \sim t_{n-k-1}$$

  a 95% confidence interval is given by

$$\widehat{\beta}_j \pm c \cdot se(\widehat{\beta}_j)$$

- The lower and the upper bounds of the confidence interval are given by

$$\underline{\beta}_j \equiv \widehat{\beta}_j - c \cdot se(\widehat{\beta}_j), \quad \overline{\beta}_j \equiv \widehat{\beta}_j + c \cdot se(\widehat{\beta}_j)$$

- **Meaning of a CI:** if random samples were obtained over and over again, with $\underline{\beta}_j$ and $\overline{\beta}_j$ computed each time, then the population value β_j would lie in the interval $(\underline{\beta}_j, \overline{\beta}_j)$ for 95% of the samples.

Example: Model of R&D Expenditures

Economists are often interested in the relationship between firm size – often measured by annual sales – and spending on research and development. Using data, we estimate the following equation

$$\widehat{\log(rd)} = \underset{(0.47)}{-4.38} + \underset{(0.060)}{1.084}\log(sales) + \underset{(0.0128)}{0.0217}profmarg, \quad n = 32, R^2 = 0.918$$

Construct a 95% confidence interval for $\beta_{\log(sales)}$.

Using Confidence Intervals for Significance

A confidence interval is a ready-made two-sided test. To test $H_0 : \beta_j = a_j$ against $H_1 : \beta_j \neq a_j$ at the 5% level:

Rule

Reject H_0 if and only if a_j lies *outside* the 95% confidence interval $[\underline{\beta}_j,\ \overline{\beta}_j]$.

- In particular, x_j is statistically significant at 5% — i.e. we reject $H_0 : \beta_j = 0$ — exactly when the 95% interval **excludes zero**.
- This is the same decision as the two-sided t-test: a_j falls outside the interval $\iff |t_{\widehat{\beta}_j}| > c$.
- The interval says *more* than the test: it also displays the whole range of values of β_j that are compatible with the data.

Testing Exclusion Restrictions

- We may want to test whether a group of variables has no effect on the dependent variable.
- The null hypothesis is that a set of variables has no effect on y, once another set of variables has been controlled.
- Consider a model with k independent variables

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + \varepsilon$$

- Suppose that we have q exclusion restrictions to test: the null hypothesis states that q of the variables have zero coefficients.

$$H_0 : \beta_{k-q+1} = 0, \ldots, \beta_k = 0$$

- The alternative hypothesis is that H_0 is not true, that is, at least one of these q coefficients is not equal to zero.

Testing Exclusion Restrictions

- When we impose the restrictions under H_0, we are left with the restricted model:

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_{k-q} x_{k-q} + \varepsilon$$

- The F-statistic (or F ratio) is given by

$$F \equiv \frac{\frac{RSS_r - RSS_{ur}}{q}}{\frac{RSS_{ur}}{n-k-1}} \sim F_{q,n-k-1}$$

- Once c has been obtained, we reject H_0 in favor of H_1 at the chosen significance level if

$$F > c$$

- If H_0 is rejected, then we say that $x_{k-q+1}, \ldots, x_k$ are jointly statistically significant at the appropriate significance level.
- If the null is not rejected, then the variables are jointly insignificant, which often justifies dropping them from the model.

Example: Major League Baseball Players' Salaries

Consider the following model

$$\log(salary) = \beta_0 + \beta_1 years + \beta_2 gamesyr + \beta_3 bavg + \beta_4 hrunsyr + \beta_5 rbisyr + \varepsilon$$

where:

- *salary* is the 1993 total salary,
- *years* is years in the league,
- *gamesyr* is average games played per year,
- *bavg* is career batting average,
- *hrunsyr* is home runs per year,
- *rbisyr* is runs batted in per year.

Suppose we want to test the null hypothesis that, once years in the league and games per year have been controlled for, the statistics measuring performance – *bavg*, *hrunsyr*, and *rbisyr* – have no effect on salary.

Example: Major League Baseball Players' Salaries

The null hypothesis is stated as

$$H_0 : \beta_3 = 0, \beta_4 = 0, \beta_5 = 0$$

The alternative is

$$H_1 : H_0 \text{ is not true.}$$

The estimated unrestricted model is

$$\widehat{\log(salary)} = \underset{(0.29)}{11.19} + \underset{(0.0121)}{0.0689}years + \underset{(0.0026)}{0.0126}gamesyr + \underset{(0.00110)}{0.00098}bavg$$
$$+\underset{(0.0161)}{0.0144}hrunsyr + \underset{(0.0072)}{0.0108}rbisyr, \quad n = 353, RSS = 183.186, R^2 = 0.6278$$

The estimated restricted model is

$$\widehat{\log(salary)} = \underset{(0.11)}{11.22} + \underset{(0.0125)}{0.0713}years + \underset{(0.0013)}{0.0202}gamesyr$$
$$n = 353, RSS = 198.311, R^2 = 0.5971$$

The R^2 Form of the F Statistic

- We can compute the F statistic using the R^2 of the models.
- Using the fact that

$$RSS_r = TSS(1 - R_r^2) \text{ and } RSS_{ur} = TSS(1 - R_{ur}^2)$$

the F statistic can be written as

$$F = \frac{\frac{R_{ur}^2 - R_r^2}{q}}{\frac{1 - R_{ur}^2}{n-k-1}} \sim F_{q,n-k-1}$$

The F Statistic for Overall Significance of a Regression

- A special set of exclusion restrictions is routinely tested.
- In the model with k independent variables, we can write H_0 in words as

$$H_0 : x_1, x_2, \ldots, x_k \text{ do not help to explain } y$$

- Stated in terms of parameters, the null is that all slope parameters are zero:

$$H_0 : \beta_1 = \beta_2 = \ldots = \beta_k = 0$$

- The alternative is that at least one of the β_j is different from zero.
- When we impose these restrictions, we get the restricted model

$$y = \beta_0 + \varepsilon$$

- The R^2 of this restricted model is zero so that the F statistic is

$$F = \frac{\frac{R^2}{k}}{\frac{1-R^2}{n-k-1}} \sim F_{k,n-k-1}$$

The Lagrange Multiplier Statistic

- The form of the *LM* statistic we derive relies on the Gauss-Markov assumptions and we do not need the normality assumption.
- To derive the *LM* statistic, consider the multiple regression model with k independent variables:

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + \varepsilon$$

- We would like to test whether, say, the last q of these variables all have zero population parameters.

$$H_0 : \beta_{k-q+1} = 0, \ldots, \beta_k = 0$$

- The *LM* statistic requires estimation of the restricted model only, thus assume we have run the regression

$$y = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \ldots + \tilde{\beta}_{k-q} x_{k-q} + \tilde{\varepsilon}$$

The Lagrange Multiplier Statistic

- If the omitted variables x_{k-q+1} through x_k truly have zero population coefficients, then, at least approximately, $\tilde{\varepsilon}$ should be uncorrelated with each of these variables in the sample.
- This suggests running a regression of these residuals on those independent variables excluded under H_0, which is almost what the LM test does.
- However, to get a usable test statistic, we must include all of the independent variables in the regression.

$$\tilde{\varepsilon} \text{ on } x_1, x_2, \ldots, x_k$$

- Under H_0, the LM statistic is given by

$$LM = n \cdot R^2 \overset{H_0}{\sim} \chi^2_q$$

where R^2 comes from the auxiliary regression.

Stata

Salary-pension Tradeoff for Teachers

Let *totcomp* denote average total annual compensation for a teacher, including salary and all fringe benefits (pension, health insurance, and so on). Write

$$totcomp = salary + benefits = salary\left(1 + \frac{benefits}{salary}\right)$$

Taking the log gives $\log(totcomp) = \log(salary) + \log(1 + b/s)$. For small b/s, $\log(1 + b/s) \approx b/s$. Let's use this approximation. This leads to the econometric model

$$\log(salary) = \beta_0 + \beta_1 \log(b/s) + \text{other factors}$$

Testing the salary-benefits tradeoff then is the same as a test of $H_0 : \beta_1 = -1$ against $H_1 : \beta_1 \neq -1$.

Estimating the Model on Stata

We use the data in MEAP93 from the datasets of Jeffrey Wooldridge. Estimating the model without any other controls gives the following output:

```
. reg lsalary bensal
```

Source	SS	df	MS	Number of obs	=	408
				F(1, 406)	=	17.05
Model	.390608607	1	.390608607	Prob > F	=	0.0000
Residual	9.30143288	406	.022909933	R-squared	=	0.0403
				Adj R-squared	=	0.0379
Total	9.69204149	407	.02381337	Root MSE	=	.15136

lsalary	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
bensal	-.8253933	.199895	-4.13	0.000	-1.218352	-.4324349
_cons	10.52318	.0415602	253.20	0.000	10.44148	10.60488

Estimating the Model on Stata

Adding Controls

Let's add some controls for school size and staff size.

```
. reg lsalary bensal lenroll lstaff droprate gradrate
```

Source	SS	df	MS		
				Number of obs	= 408
				F(5, 402)	= 45.43
Model	3.49912092	5	.699824185	Prob > F	= 0.0000
Residual	6.19292056	402	.015405275	R-squared	= 0.3610
				Adj R-squared	= 0.3531
Total	9.69204149	407	.02381337	Root MSE	= .12412

lsalary	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
bensal	-.5893175	.1648739	-3.57	0.000	-.9134402	-.2651948
lenroll	.0881206	.007324	12.03	0.000	.0737224	.1025187
lstaff	-.2182771	.0499504	-4.37	0.000	-.3164737	-.1200806
droprate	-.0002826	.0016145	-0.18	0.861	-.0034565	.0028913
gradrate	.0009674	.0006625	1.46	0.145	-.0003351	.0022699
_cons	10.73846	.2582652	41.58	0.000	10.23074	11.24618

Reporting Regression Results

	(1) lsalary	(2) lsalary	(3) lsalary
bensal	-0.8254***	-0.6048***	-0.5893***
	(0.1999)	(0.1654)	(0.1649)
lenroll		0.0874***	0.0881***
		(0.0073)	(0.0073)
lstaff		-0.2220***	-0.2183***
		(0.0501)	(0.0500)
droprate			-0.0003
			(0.0016)
gradrate			0.0010
			(0.0007)
Constant	10.5232***	10.8438***	10.7385***
	(0.0416)	(0.2516)	(0.2583)
Observations	408	408	408
R-squared	0.040	0.353	0.361

Standard errors in parentheses
* p<0.10, ** p<0.05, *** p<0.01

Exercise

Rent Rates and Student Population

Are rent rates influenced by the student population in a college town? Let *rent* be the average monthly rent paid on rental units in a college town in the United States. Let *pop* denote the total city population, *avginc* the average city income, and *pctstu* the student population as a percentage of the total population. One model to test for a relationship is

$$\log(rent) = \beta_0 + \beta_1 \log(pop) + \beta_2 \log(avginc) + \beta_3 pctstu + \varepsilon$$

Rent Rates and Student Population

We estimate the model using 1990 data from RENTAL for 64 college towns.

```
. reg lrent lpop lavginc pctstu
```

Source	SS	df	MS	Number of obs	=	64
				F(3, 60)	=	16.89
Model	1.15895493	3	.38631831	Prob > F	=	0.0000
Residual	1.37204467	60	.022867411	R-squared	=	0.4579
				Adj R-squared	=	0.4308
Total	2.5309996	63	.040174597	Root MSE	=	.15122

lrent	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
lpop	.065868	.038826	1.70	0.095	-.0117956	.1435316
lavginc	.5070147	.0808357	6.27	0.000	.3453193	.6687101
pctstu	.0056297	.0017421	3.23	0.002	.002145	.0091144
_cons	.0427819	.8438762	0.05	0.960	-1.645222	1.730786

References

[1] Wooldridge, J. M. (2019). *Introductory Econometrics: A Modern Approach* (7th ed.). Cengage Learning.

[2] Stock, J. H., & Watson, M. W. (2019). *Introduction to Econometrics* (4th ed.). Pearson.

[3] Angrist, J. D., & Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.